C1 Atomic Structure and the Periodic Table – Knowledge Organiser

AQA GCSE Chemistry • Set 1: Atoms, Elements and Mixtures **Paper 1**

#	Question	Answer
1	Atom	Smallest part of an element that can exist
2	Element	Substance made of only one type of atom
3	Compound	Two or more elements chemically combined in fixed proportions
4	Mixture	Two or more substances not chemically combined
5	Methods to separate mixtures	Filtration, crystallisation, distillation, chromatography
6	Radius of an atom	About 0.1 nm (1×10^{-10} m)
7	Size of the nucleus compared with the atom	About 1/10 000 of the radius
8	Relative masses of proton, neutron, electron	1, 1, very small
9	Relative charges of proton, neutron, electron	+1, 0, −1
10	Why atoms have no overall charge	Equal numbers of protons and electrons

C1 Atomic Structure and the Periodic Table – Knowledge Organiser

AQA GCSE Chemistry • Set 2: Atomic Number, Isotopes and the Atomic Model **Paper 1**

#	Question	Answer
1	Atomic number	Number of protons
2	Mass number	Number of protons + neutrons
3	Isotopes	Same number of protons, different number of neutrons
4	Relative atomic mass	Average mass of an element's atoms, taking isotope abundance into account
5	Plum pudding model	Ball of positive charge with electrons embedded in it
6	Experiment that replaced the plum pudding model	Alpha particle scattering
7	Conclusion of alpha scattering	Mass is concentrated in a tiny positive nucleus
8	Scientist who put electrons in shells	Niels Bohr
9	Scientist who showed neutrons exist	James Chadwick
10	Maximum electrons in the first three shells	2, 8, 8

C1 Atomic Structure and the Periodic Table – Knowledge Organiser

AQA GCSE Chemistry • Set 3: The Periodic Table **Paper 1**

#	Question	Answer
1	Modern periodic table is ordered by	Atomic number
2	Group number tells you	Number of outer electrons
3	Period number tells you	Number of electron shells
4	Ions formed by metals	Positive ions
5	Ions formed by non-metals	Negative ions
6	Position of metals in the periodic table	Left and centre
7	Mendeleev ordered elements by	Atomic weight and properties
8	Why Mendeleev left gaps	For undiscovered elements
9	Why Mendeleev swapped some pairs	To match their properties
10	What explained the swapped pairs	Isotopes

C1 Atomic Structure and the Periodic Table – Knowledge Organiser

AQA GCSE Chemistry • Set 4: Groups 0, 1 and 7 **Paper 1**

#	Question	Answer
1	Name of Group 0	Noble gases
2	Why noble gases are unreactive	Full outer shell (stable)
3	Trend in boiling point down Group 0	Increases
4	Name of Group 1	Alkali metals
5	Trend in reactivity down Group 1	Increases
6	Products of a Group 1 metal with water	Metal hydroxide + hydrogen
7	Equation for sodium with water	$2Na + 2H_2O \rightarrow 2NaOH + H_2$
8	Name of Group 7	Halogens
9	Trend in reactivity down Group 7	Decreases
10	Trend in melting and boiling point down Group 7	Increases

C1 Atomic Structure and the Periodic Table – Knowledge Organiser

AQA GCSE Chemistry • Set 5: Group 7 and Reactivity **Paper 1**

#	Question	Answer
1	Form halogens exist in	Diatomic molecules
2	Appearance of chlorine	Green gas
3	Appearance of bromine	Red-brown liquid
4	Appearance of iodine	Grey-black solid
5	Halogen displacement	More reactive halogen displaces a less reactive one from its salt
6	Products of chlorine + potassium bromide	Potassium chloride + bromine
7	Charge on Group 1 ions	1+
8	Charge on Group 7 ions	1−
9	Why Group 1 gets more reactive down the group	Outer electron further from nucleus, lost more easily
10	Why Group 7 gets less reactive down the group	Outer shell further from nucleus, electron gained less easily

C1 Atomic Structure and the Periodic Table – Knowledge Organiser

AQA GCSE Chemistry • Set 6: Transition Metals Paper 1 Triple only

#	Question	Answer
1	Position of transition metals	Central block, between Groups 2 and 3
2	Transition metals compared with Group 1	Harder, denser, higher melting point, less reactive
3	Examples of transition metals	Cr, Mn, Fe, Co, Ni, Cu
4	Example of ions with different charges	Fe^{2+} and Fe^{3+}
5	Appearance of transition metal compounds	Coloured
6	Common use of transition metals	Catalysts
7	Catalyst in the Haber process	Iron
8	Colour of copper(II) compounds	Blue
9	Colour of iron(II) compounds	Green
10	Colour of iron(III) compounds	Orange-brown